

Computer Vision with Deep Learning

YOLO detection

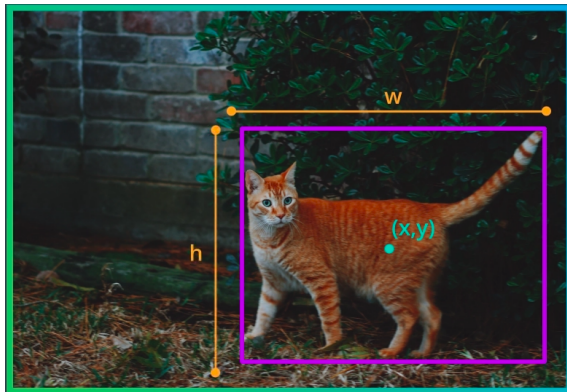
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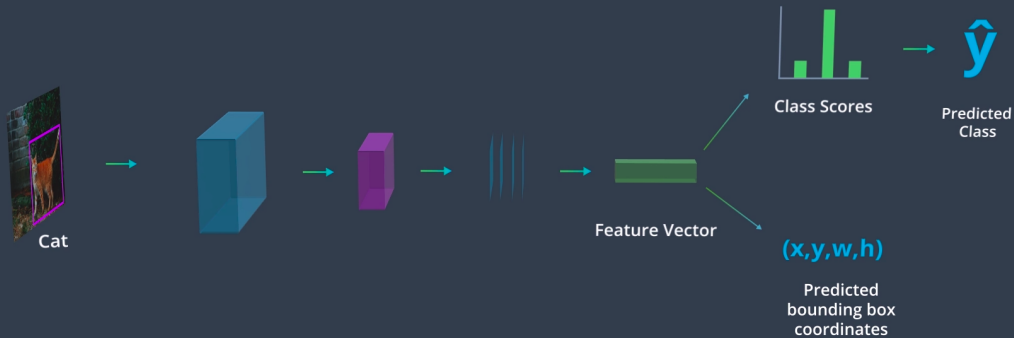
Object detection

- Object detection vs classification



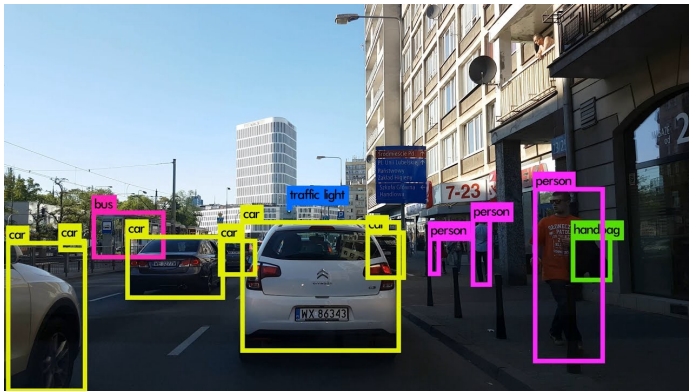
Object detection

Using the same architecture



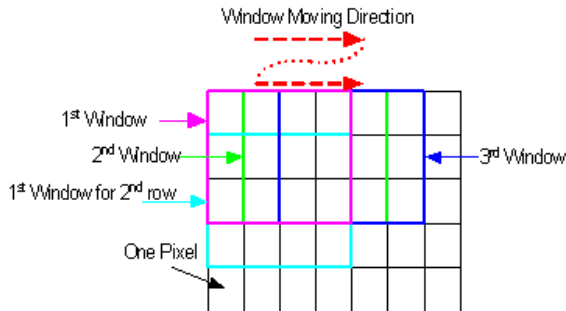
Multiple objects

- How can we detect multiple objects?



Multiple objects

- Sliding window ¹

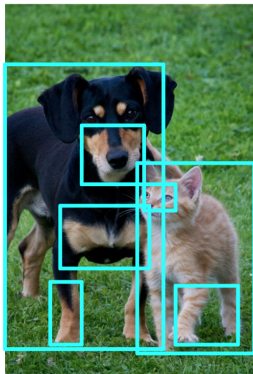


- Extremely expensive

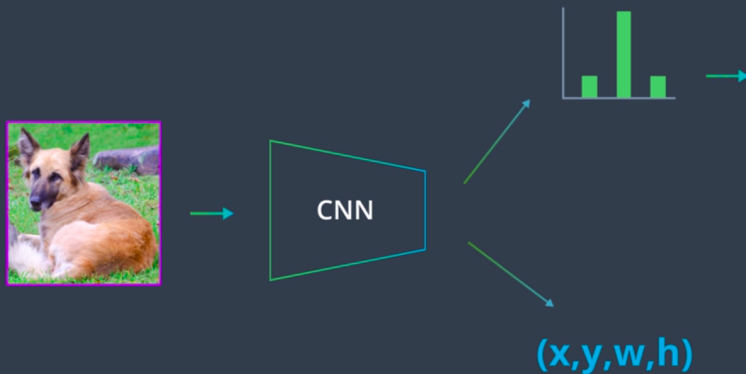
¹Felzenszwalb, Pedro F., et al. "Object detection with discriminatively trained part-based models." IEEE transactions on pattern analysis and machine intelligence 32.9 (2009): 1627-1645.

Region proposals

- Regions only for areas which might contain an object. ²
- We may use boundaries, regions, texture, feature points... etc.



²Girshick, Ross, et al. "Rich feature hierarchies for accurate object detection and semantic segmentation." Proceedings of the IEEE conference on computer vision and pattern recognition. 2014.



- input: proposed region; output: label or background

You Only Look Once: Unified, Real-Time Object Detection

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<http://pjreddie.com/yolo/>

Abstract

We present YOLO, a new approach to object detection. Prior work on object detection repurposes classifiers to perform detection. Instead, we frame object detection as a regression problem to spatially separated bounding boxes and associated class probabilities. A single neural network predicts bounding boxes and class probabilities directly from full images in one evaluation. Since the whole detection pipeline is a single network, it can be optimized end-to-end directly on detection performance.

Our unified architecture is extremely fast. Our base YOLO model processes images in real-time at 45 frames per second. A smaller version of the network, Fast YOLO, processes an astounding 155 frames per second while still achieving double the mAP of other real-time detectors. Compared to state-of-the-art detection systems, YOLO makes more localization errors but is less likely to predict false positives on background. Finally, YOLO learns very general representations of objects. It outperforms other detection methods, including DPM and R-CNN, when generalizing from natural images to other domains like artwork.

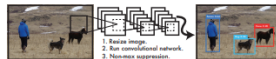


Figure 1: The YOLO Detection System. Processing images with YOLO is simple and straightforward. Our system (1) resizes the input image to 448×448 , (2) runs a single convolutional network on the image, and (3) thresholds the resulting detections by the model's confidence.

methods to first generate potential bounding boxes in an image and then run a classifier on these proposed boxes. After classification, post-processing is used to refine the bounding boxes, eliminate duplicate detections, and rescore the boxes based on other objects in the scene [13]. These complex pipelines are slow and hard to optimize because each individual component must be trained separately.

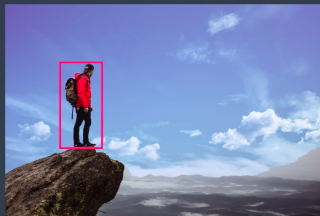
We reframe object detection as a single regression problem, straight from image pixels to bounding box coordinates and class probabilities. Using our system, you only

³Redmon, Joseph, et al. "You only look once: Unified, real-time object detection." Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.

YOLO preliminaries

Output

- The output should be a vector



CNN



$$y = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \end{bmatrix}$$

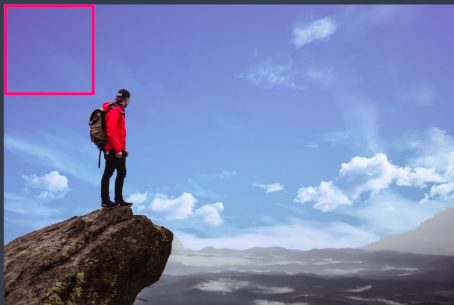
$c_1 = person$

$c_2 = cat$

$c_3 = dog$

Sliding window

- p_c outputs the probability that an object is in the window.
- It works great but it is expensive.

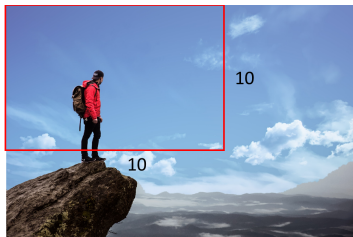


$$y = \begin{bmatrix} p_c \\ c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \end{bmatrix}$$

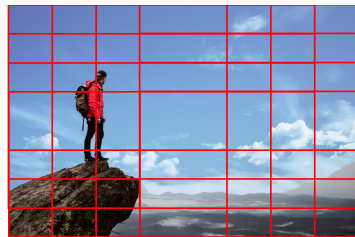
$c_1 = \text{person}$
 $c_2 = \text{cat}$
 $c_3 = \text{dog}$

Analyzing sliding windows

- Sliding window



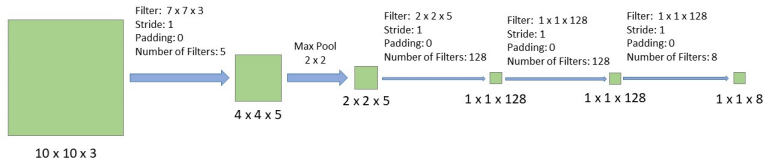
16 x 16 x 3



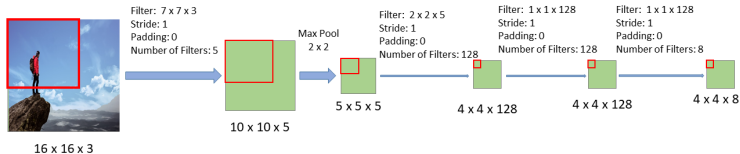
16 x 16 x 3

Analyzing sliding windows

- Passing the window through a CNN

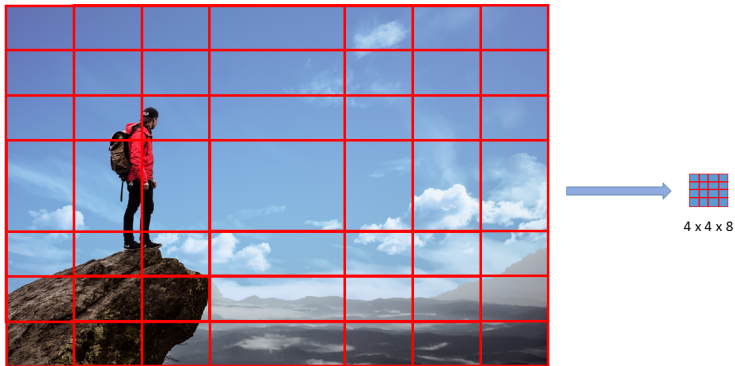


- Whole image through the CNN



Eureka!

- The windows are represented in the last layer of the CNN!



A grid instead windows

Improving localization with cells



Associated Vectors

1. If an object is in that cell.
2. The "class" of that object.
3. The predicted bounding box for that object.

A grid instead windows

Improving localization



$c_1 = person$ $c_2 = cat$ $c_3 = dog$

$$g_n = \begin{bmatrix} p_c \\ c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \end{bmatrix}$$

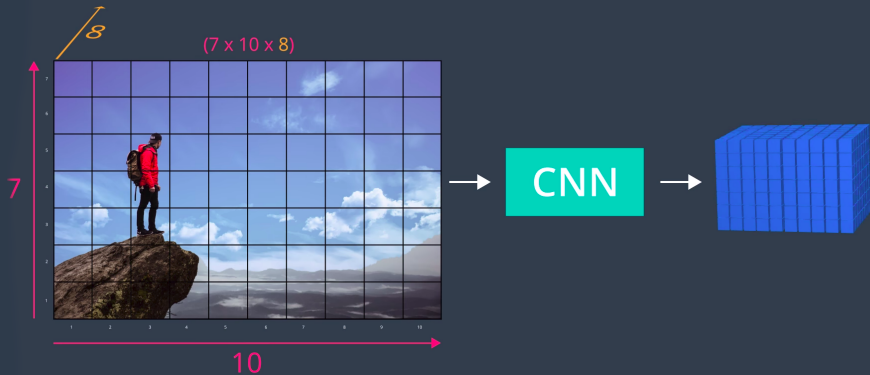
Training labels



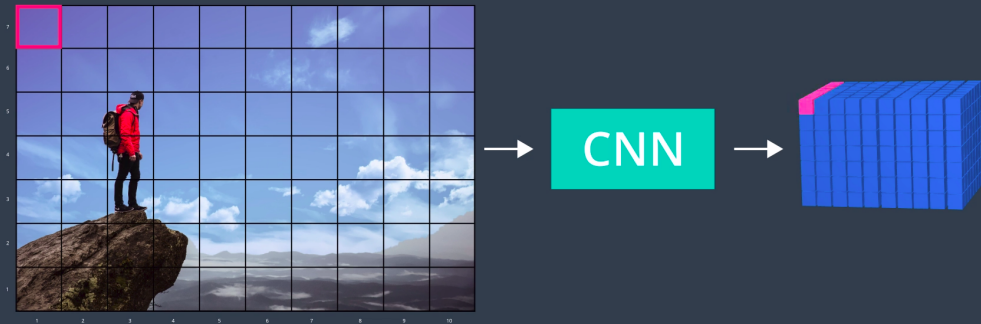
For each
training image

- Break into a grid
- Manually assign a ground truth vector to each grid cell

Training labels

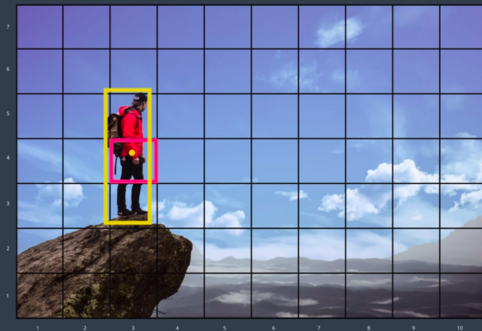


Training labels



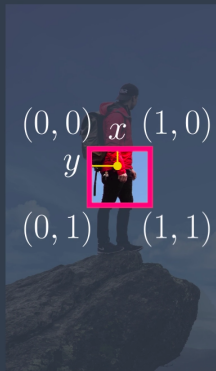
Training labels

- Training locations.
- The location of one object is assign to only one grid cell.



Training labels

- The center of the object is relative to the cell.



$$g_{33} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0.5 \\ 0.3 \\ 0.1 \\ 0.4 \end{bmatrix}$$

Training labels

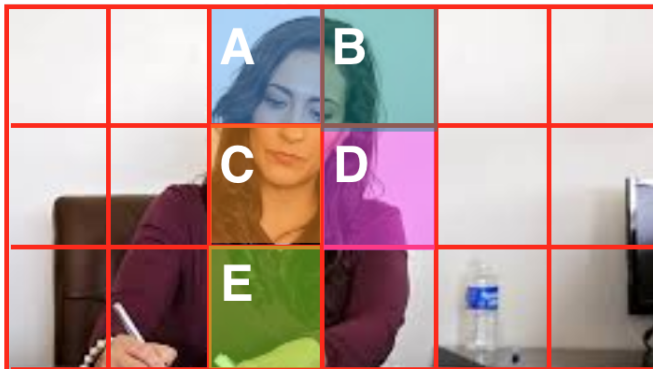
- The width and height are relative to the image.



$$g_{33} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0.5 \\ 0.3 \\ 0.1 \\ 0.4 \end{bmatrix} \begin{matrix} x \\ y \\ w \\ h \end{matrix}$$

Quiz

- Which cell will provide the correct label?



Multiple boxes

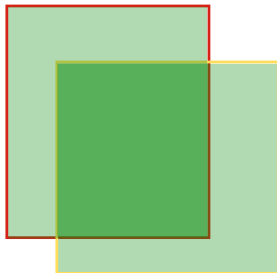
- A problem with grid cells is that they provide too many boxes.



Non Maximum suppression

- Intersection over union

$$IOU = \frac{A_{intersection}}{A_{union}} \quad (1)$$



- What is the range of the function?

Non Maximum suppression

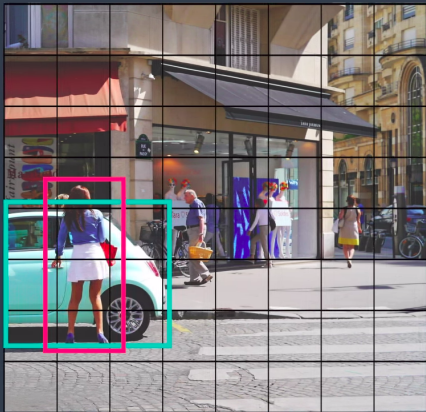
- 1 Look at the output vector of each grid cell.
- 2 Remove all bounding boxes that have a p_c value less than or equal to some threshold.

$$p_c < 0.5$$

- 3 Select the bounding box with the highest p_c value.
- 4 Remove all the bounding boxes that have a high IoU with the box selected in the last step.

Anchor boxes

- Same center problem.

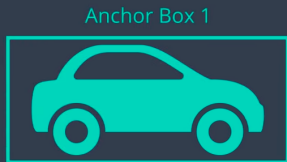


$$y = \begin{bmatrix} p_c \\ c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \end{bmatrix}$$

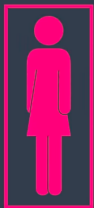
$c_1 = person$
 $c_2 = car$
 $c_3 = dog$

Anchor boxes

- Each anchor box as different aspect ratio.



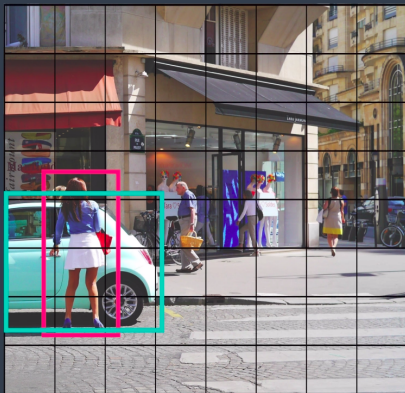
Anchor Box 2



$$y = \begin{bmatrix} p_c \\ c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \\ p_c \\ c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \end{bmatrix}$$

Anchor boxes

- We can place as many anchor boxes as we want.



$$y = \begin{bmatrix} p_c \\ c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \\ p_c \\ c_1 \\ c_2 \\ c_3 \\ x \\ y \\ w \\ h \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ x \\ y \\ w \\ h \\ 1 \\ 1 \\ 0 \\ 0 \\ x \\ y \\ w \\ h \end{bmatrix}$$

$c_1 = \text{person}$
 $c_2 = \text{car}$
 $c_3 = \text{dog}$

YOLO Architecture

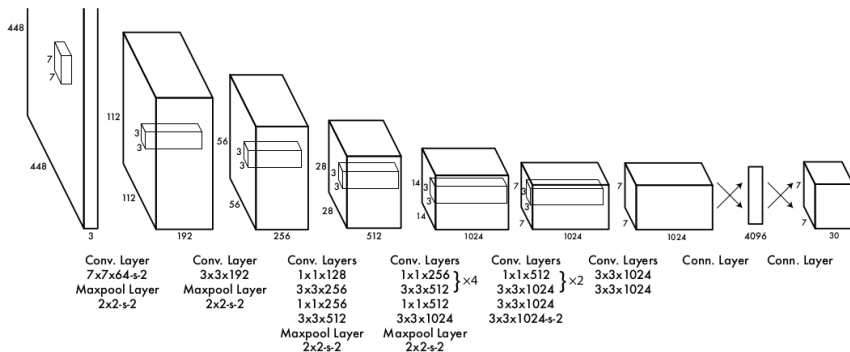


Figure 3: The Architecture. Our detection network has 24 convolutional layers followed by 2 fully connected layers. Alternating 1×1 convolutional layers reduce the features space from preceding layers. We pretrain the convolutional layers on the ImageNet classification task at half the resolution (224×224 input image) and then double the resolution for detection.

YOLO Results

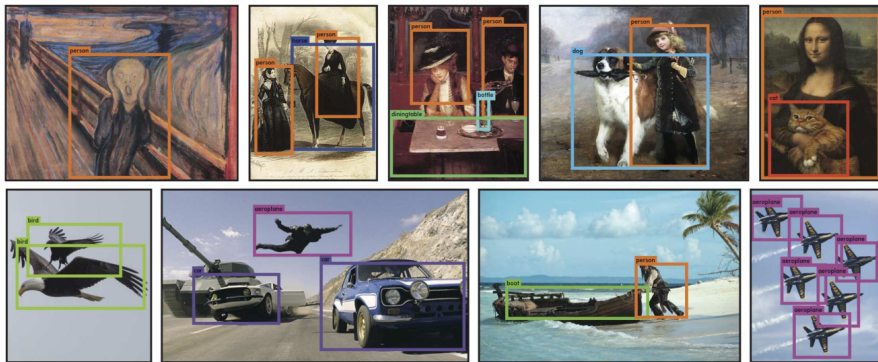


Figure 6: Qualitative Results. YOLO running on sample artwork and natural images from the internet. It is mostly accurate although it does think one person is an airplane.

- Redmon, Joseph, et al. "You only look once: Unified, real-time object detection." Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.
- Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1). Cambridge: MIT press.
- Alex Smola and S.V.N. Vishwanathan, Introduction to Machine Learning, Cambridge University Press.
- Udacity, Computer vision nanodegree